Case Management Workbench: giving Maria her clinical judgment back

HiLabs APM case | MVP for a Medicare Advantage / Medicaid health plan | Deck + Excel annex (5 synthetic patients, live models)

The answer in 30 seconds

1. Problem. An RN with 48 cases loses ~69% of a 480-min day to hunting data, re-typing documentation, redialling bad numbers and triaging her own queue. Only ~12% is real conversation.

2. Product. A web workbench with three screens: **Today queue** (who, why now, what next), **Member 360** (one trustworthy picture: every fact has a source, an as-of date and a conflict flag), **Outreach & documentation** (best number/time/channel; notes and assessments pre-filled).

3. Principle. AI drafts and ranks; the nurse confirms and decides. Nothing is auto-sent, auto-diagnosed or auto-closed.

69% — of Maria's day lost to data hunting, documentation, outreach friction and triage (model, medium confidence)

12% > 39% — share of day in real conversations: baseline vs design model (pilot commitment: >= 25%)

21 / 5 — datasets inventoried / fictional patients, each with 8-10 records and planted imperfections

A Understand
Persona, day in the life, ranked pain, where time goes

B Data strategy
21 datasets, each tied to a decision; 5 insight recipes

C Synthetic patients
5 scenarios, imperfections that break naive designs

D Product design
Queue, Member 360, Outreach/Doc; AI vs human

E Prioritize & measure
MVP cut, metrics w/ baselines, risks, pilot & pricing

What I would build first, and what I deliberately leave out

Build first (MVP).

- ADT-triggered **queue with a clock** (2-business-day contact, 7-day follow-up) and a transparent score
- **Member 360** on 9 must-have datasets with freshness banners and side-by-side medication conflicts
- **Contact graph**: rank phone numbers by source, recency and past success; recommend channel and time
- **Pre-filled assessment and note draft** with citations; one confirm click

Leave out (on purpose).

- Autonomous outreach or auto-closing cases: trust and compliance risk too high for v1
- Diagnosis or treatment suggestions: we flag conflicts for the clinician, never resolve them
- Rebuilding the EHR or claims portal: link out, do not replicate
- Member-facing app, wearables, Medicaid LTSS feeds: Phase 2 / Later
- More predictive models: the data-assembly gap is bigger than the prediction gap

The test I applied to every item: what would Maria do differently because she saw this? Example: Eleanor Marsh (HF, discharged Fri) - *discharge alert* + *new-drug pharmacy reject* + *PA status* tells Maria her new heart-failure drug was never dispensed and nobody has filed the PA. She calls today, instead of discovering it on day 7 or after a readmission (~1 in 5 Medicare discharges is readmitted within 30 days).

Sources: HiLabs case brief; Jencks et al. NEJM 2009 (30-day readmission 19.6%); CMS TCM 99495/99496. Full source list and assumptions in annex sheet 'Assumptions_Sources'.

A. Understand: Maria is the best clinical sensor in the plan, and her day is spent on plumbing

Persona (case manager only) and a day in her life. Her instinct about a med list that does not add up is the scarce asset.

MR

Maria R., RN, 20 yrs ICU

Complex case manager at a Medicare Advantage plan; telephonic; 48 active cases

Job to be done
Get each member safely through a transition: right meds, right follow-up, right services, before the next ED visit.

Strengths
Clinical pattern-recognition; rapport; knows the community resources.

Needs from software
1) Tell me who needs me today and why. 2) Show one reliable picture, with how old each fact is. 3) Get me to a connected call. 4) Do not make me type it twice.

Success looks like
Contact within 2 business days, med reconciliation done, visit within 7 days, no preventable readmission.

Fears
Missing a deadline; trusting wrong data; an audit finding.

Monday in Maria's life (from the case brief; times illustrative)

- **8:00** 6 discharge alerts since Fri; claims show nothing yet; picks a case by gut feel
- **8:20** Opens claims portal, auth system, pharmacy file, HIE, old notes, own spreadsheet for ONE member
- **9:10** Phone on file is disconnected; searches three systems for another number
- **9:40** Latest claim is 45 days old; alert says 'admitted', not why
- **10:30** Two systems disagree on meds (duplicate statin? stopped ACE-inhibitor?)
- **11:15** 40-field assessment to close the contact; same facts re-typed into plan and audit form
- **13:30** Three voicemails; caregiver number is buried in a free-text note
- **15:00** Two real conversations so far today. Instinct barely used.
- **17:30** Charting after hours (assumed ~45 min)

Opportunity: every red dot above is a data or workflow problem, not a clinical one.

What a bad day costs: six members with a 2-business-day clock, of whom Maria reaches ~2 by mid-afternoon; every unreached discharge is a lost chance to catch a missing medication before it becomes a readmission, and a missed HEDIS TRC / TCM window for the plan.

Why this persona only

Per the brief, the case manager is the user. Supervisors, nurses on other programs and members matter, but a queue that works for Maria also gives supervisors visible capacity and gives members faster contact. I model Maria at 48 cases (the complex-case range is 30-60); the transitions-of-care nurse at 60-100+ cases feels the same pain with shorter cycles, so the design scales to her.

Sources: HiLabs case brief (Maria scenario, workload ranges). Times other than those quoted in the brief are illustrative; after-hours charting is an assumption to validate.

A. Understand: ~69% of her day goes to data, typing, dialling and triage; ~12% is real conversation

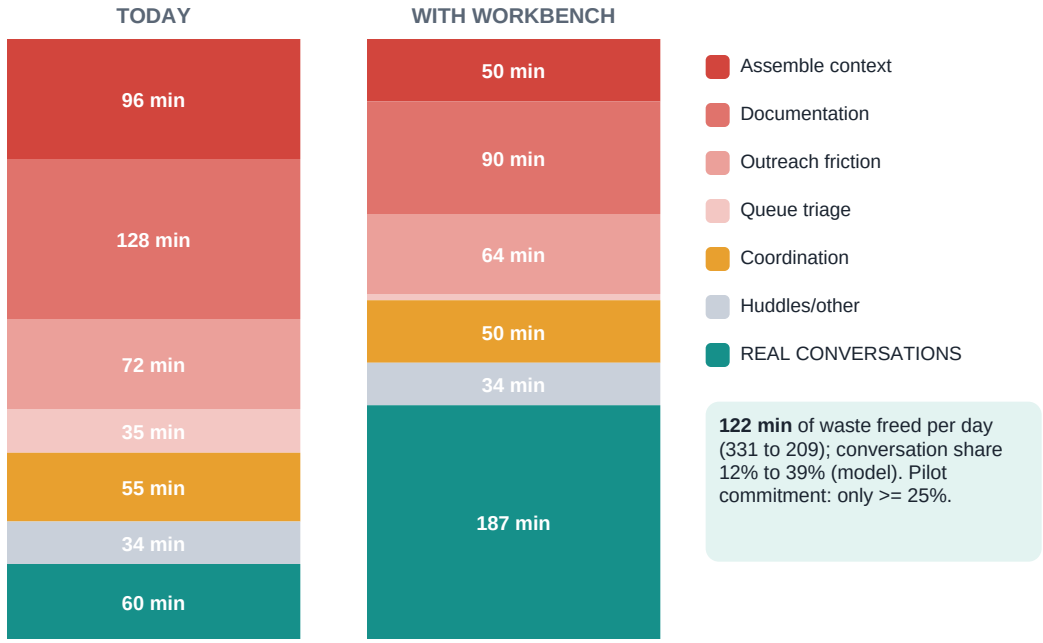
Pain points ranked by impact (minutes lost + clinical risk) and a time model with assumptions shown. Live version in annex sheet 'Time_Model'.

Pain points ranked by impact

#	Pain point	Min/day	Clinical risk	Evidence
1	Documentation re-typed across assessment, plan, notes, audit forms	128	Medium	Case brief; nurse time studies: ~19-27% of time on documentation
2	Context assembly across 6-7 screens	96	High	Case brief; missed signals = missed med/safety issues
3	Hard outreach: dead numbers, voicemail, wrong channel/time	72	High	Case brief; no contact = no intervention
4	Stale / delayed data hides events (claims 45d, ADT without reason)	indirect	High	Claims run-out 30-90d; ADT has no diagnosis
5	Conflicting meds / diagnoses across sources	indirect	Highest	Safety: duplicate or contraindicated therapy
6	Unprioritised queue, caseload spikes	35	Medium	Case brief; 6 alerts over a weekend

Ranking logic. Impact = minutes lost x frequency + clinical risk of the failure. Documentation costs the most minutes; **conflicting meds** costs the fewest minutes but the most harm, so it gets a first-class design treatment (side-by-side, never silently merged). Stale data and conflicts are 'indirect': they do not take minutes, they take correctness.

Where 480 minutes go: today vs with workbench (model)



122 min of waste freed per day (331 to 209); conversation share 12% to 39% (model). Pilot commitment: only >= 25%.

How I got the numbers, and how sure I am

Activity	Today formula	Assumption and basis	Confidence
Assemble context	6.0 min x 16 touches = 96	~1 min per screen x 6-7 systems (brief); 16 touches = about a third of 48 cases + 6 new alerts	Medium
Documentation	8.0 min x 16 touches = 128	Call note ~5 min; full 40-field assessment ~15 min on ~1 in 3 touches; = 27% of day, top of the 19-27% range seen in nurse time studies (RAND 2012; Baumann 2018); case management is phone + paperwork, so I assume the high end	Medium
Outreach friction	4.5 min x 16 = 72	~2.5 attempts per touch x ~1.8 min lookup/dial/voicemail; ~30% attempts connect (assumption)	Low-Medium
Triage / spreadsheet	35 min fixed	Morning queue scan + after each alert wave	Medium
Conversations	remainder = 60 min	Cross-check: ~4 connected calls x 15 min = 60 min; matches '2 real conversations by 3 pm'	Medium

Sensitivity: if documentation is 6 min/touch instead of 8, waste share is still ~62%. Targets assume 20 touches/day with the workbench and that the nurse confirms every AI draft. To be validated by shadowing 5-8 nurses (time-and-motion) in week 1 of the pilot.

Sources: HiLabs case brief; Hendrich/RAND 2012 (nurses ~19% of time on documentation); Baumann et al. Health Policy 2018 (nurses 23% post-EHR); Sinsky et al. Ann Intern Med 2016. All per-touch minutes are my estimates, flagged for pilot validation.

B. Data strategy (1/2): triggers, clinical truth and reachability are the must-haves

Every dataset is tied to a decision. MVP = must-have for the first release; Phase 2 / Later = valuable but costlier or lower frequency. Full 21-row table in annex.

Dataset	Source	What it tells Maria -> decision it enables	Freshness	Expected quality problem -> how handled	Priority
D1 Member & eligibility	Plan enrollment (834), MDM/EMPI	Who, plan/line of business, PCP, language, dual status, coverage end date **-> Route case by language/licence; warn if coverage is about to lapse**	Daily	Retro-terms, dup IDs -> EMPI golden ID; show coverage-end warning	MVP
D2 Hospital alerts (ADT)	HL7 ADT via HIE / alert vendor / hospital feeds	Admit, ED, discharge, transfer; facility; disposition **-> Trigger the case; start the 2-business-day clock**	Minutes-hours	No diagnosis, out of order, dup alerts, name variants -> dedupe by encounter; show match confidence; infer reason from D5/D3 later	MVP
D3 Medical claims	Plan claims warehouse	Confirmed history: visits, procedures, dx codes, costs, Z-codes **-> Confirm utilisation pattern (ED x6), find open gaps**	30-90 day lag	Lag, reversals, adjustments -> use as history/confirmation, never as the trigger; label 'data through' date	MVP
D4 Pharmacy fills & rejects	PBM / pharmacy claims (incl. rejected claims)	Fills, days supply, PDC adherence, rejected claims (PA needed), prescriber, pharmacy **-> Has the member started the new drug? Why not?**	Daily-weekly	Cash / discount-card fills invisible, mail order -> show confidence; compute PDC; treat reject as a signal	MVP
D5 Clinical records (discharge summary, CCD/FHIR)	EHR / HIE (C-CDA, FHIR)	Why admitted, discharge meds & instructions, problem list, vitals, EF, weight **-> Reason for stay; discharge med list; red-flag symptoms**	Hours-days, partial	Missing/partial docs, free text -> NLP extraction with citations; show 'not received yet'	MVP
D6 Labs & results	Lab feed / HIE (LOINC)	A1c, eGFR, K, BNP, INR trends **-> Is disease controlled? Safe to escalate to prescriber?**	Days	Unmapped LOINC, unit mismatch, delay -> normalise units; show trend + collected date	MVP-lite
D7 Authorizations / UM	UM system	Pending/denied home health, DME, SNF, drug PAs **-> Unblock the service that is stuck; submit PA**	Daily	Status codes differ by vendor -> map to 5 states	MVP
D8 Appointments & visit status	Provider scheduling / EHR; claims as fallback	Booked, kept, no-show, days to follow-up visit **-> Is the 7-day follow-up booked and inside the window?**	Hours-days (feeds) / lag (claims)	Few plans get scheduling feeds -> start with claims + HIE encounters, mark 'unknown' honestly	MVP-lite
D9 Contact & outreach history	CM system, call centre, telephony, SMS	Every attempt, outcome, channel, time, who answered **-> Stop redialling dead numbers; pick channel and time**	Real time	Free text, inconsistent dispositions -> structured disposition codes	MVP
D10 Contact graph (phones, address, caregiver, consent)	Enrollment, pharmacy, provider, claims, call centre, line-type validation	Ranked phone numbers with source/last-verified/success rate; caregiver; best time; SMS/TCPA consent **-> Whom to call, when, on which number**	Continuous	Conflicting numbers, landlines, caregiver buried in notes -> rank by recency x verified success; NLP lifts numbers from notes	MVP
D11 Case, care plan & assessment history	Existing CM platform, legacy spreadsheets	Prior goals, barriers, assessments, closed-unable-to-reach **-> Do not re-ask; build on last plan**	Real time	Dup cases, spreadsheets -> import + merge candidates; carry forward unchanged answers	MVP

Sources: ADT/HIE alert vendors (Collective Medical, PointClick), HL7 v2 ADT, C-CDA/FHIR; NCQA HEDIS; AHC-HRSN / PRAPARE; 42 CFR Part 2; TCPA. Freshness figures are typical industry ranges, not plan-specific measurements.

B. Data strategy (2/2): prioritisation, social context and capacity round out the picture

Every dataset is tied to a decision. MVP = must-have for the first release; Phase 2 / Later = valuable but costlier or lower frequency. Full 21-row table in annex.

Dataset	Source	What it tells Maria -> decision it enables	Freshness	Expected quality problem -> how handled	Priority
D12 Risk & priority scores	HiLabs / plan models (HCC, readmission, ED propensity, rising risk)	Predicted risk, trend, drivers **-> Whom to work first when everything looks urgent**	Weekly-daily	Black-box -> show top 3 drivers; calibrate per plan	MVP
D13 Behavioral health data	BH claims, facility feeds, PHQ-9/GAD-7	BH admits, meds, screens; access restrictions **-> Enforce 7-day post-psych follow-up**	Days-weeks	42 CFR Part 2 / consent limits -> role-based masking, consent flag	MVP (non-SUD)
D14 SDOH screens & signals	HRA, AHC-HRSN/PRAPARE, Z-codes, call-centre notes, ADI index	Housing, food, transport, isolation **-> Which barrier to solve first**	Months old	Stale self-report -> age badge; newer signals override with evidence	Phase 2 (basic in MVP)
D15 Quality / care gaps	HEDIS engine (TRC, FUH, FMC, eye exam, A1c, statin)	Open measures and deadlines **-> Close a gap while on the call**	Daily-monthly	Measure logic changes yearly -> versioned rules	Phase 2
D16 Provider & network directory	Plan directory, FQHC / home-health lists	In-network PCP/specialist/home-health, location, hours, telehealth **-> Book a reachable, in-network follow-up**	Weekly	Stale directories -> 'verified on' date	Phase 2
D17 Benefits & supplemental	Benefit config	Rides, meals, OTC, respite, DME limits **-> Offer the benefit that solves the barrier**	Plan-year	Per-plan variants -> rule table	Phase 2
D18 Caregiver & advance directives	Assessments, notes, state registries	Caregiver identity, burden score, authorised rep, directives **-> Who can I talk to, and is the caregiver at risk?**	Months	Mostly in free text -> NLP + confirm	Phase 2
D19 Medicaid / LTSS (dual-eligible)	State Medicaid feeds	Authorised vs delivered personal-care hours, redetermination dates **-> Escalate under-delivered home care**	Monthly	Slow, state formats -> monthly freshness badge	Later
D20 Remote monitoring / patient-reported	RPM vendors, wearables, member app	Daily weights, BP, glucose **-> Catch decompensation before the ED**	Daily	Sparse, non-standard -> only for enrolled programs	Later
D21 CM capacity & skills	HR / workforce, CM system	Licensure state, languages, caseload, calendar, program certification **-> Route the right case to the right nurse**	Real time	Self-reported skills -> manager-owned table	MVP

Design rule for data trust. Every value in the UI carries **source + as-of date**; every claim-derived fact carries a **'data through'** banner; conflicts show **both values** and who said what; an event not yet visible is drawn as a **ghost item ('not received yet')**, never as absence. ADT triggers; claims confirm; pharmacy and clinical records explain.

Sources: ADT/HIE alert vendors (Collective Medical, PointClick), HL7 v2 ADT, C-CDA/FHIR; NCQA HEDIS; AHC-HRSN / PRAPARE; 42 CFR Part 2; TCPA. Freshness figures are typical industry ranges, not plan-specific measurements.

B. Data strategy: value comes from combinations, not single feeds

Five 'insight recipes' that the engine computes and the queue shows as a one-line reason. Each ends in an action Maria would not otherwise have taken today.

	INPUTS		WHAT THE NURSE SEES	WHAT SHE DOES DIFFERENTLY
Heart-failure drug not started	ADT discharge (D2) + discharge summary new drug (D5) + pharmacy REJECT (D4) + UM PA status (D7)	>	New HF medication was rejected for prior auth and nobody has submitted the PA	**Submit PA today; call prescriber; confirm what she is actually taking**
Silent adherence gap	Pharmacy fills (D4) + labs (D6) + risk trend (D12) + contact history (D9)	>	A1c rose to 9.4 and metformin PDC is 54%: the gap is real, the member has gone quiet	**Ask about cost/side effects, not 'are you compliant?'**
Frequent ED, wrong number	ADT (D2) + claims (D3) + contact graph (D10) + contact history (D9)	>	ADT shows 6 ED visits (claims show 4). 2 of 3 numbers are dead; newest pharmacy number never tried	**Call number #3 and text; ask PCP/pharmacy to relay**
Follow-up outside the window	ADT discharge (D2) + BH data (D13) + appointments (D8) + SDOH (D14)	>	Psych discharge day 6, follow-up booked day 8, no meds picked up, housing record is stale	**Rebook <=7 days by telehealth; text first; send Rx to nearby pharmacy**
Polypharmacy + caregiver strain	Pharmacy, 2 pharmacies (D4) + labs (D6) + caregiver (D18) + LTSS hours (D19) + ADT fall (D2)	>	Duplicate statins, a Beers-list sleep drug, A1c 6.3 on a sulfonylurea, ED fall, caregiver burden high, home-care hours under-delivered	**Pharmacist review; escalate agency; respite benefit; call daughter in the evening**

How an insight is built (so it is auditable): rule or model fires > evidence list (dataset, record, as-of date) > confidence (High: confirmed in 2 sources; Medium: one source or may be missing cash fills; Low: inferred) > suggested action.
Example of explicit uncertainty: 'No metformin fill since 12 Jun (PDC 54%) - confidence Medium: cash or discount-card fills are invisible. Ask: where do you fill it?'
Deliberately not built: a generic 'risk score' tile with no reason. A number without a reason is not actionable and erodes trust.

Sources: NCQA HEDIS TRC/FUH/FMC; CMS Part D adherence (PDC >= 80%); AGS Beers Criteria; Cutler et al. BMJ Open 2018 (non-adherence cost).

C. Five synthetic patients, each planted with imperfections that break a naive design

Each has 8-10 records across datasets (annex sheet 'Synthetic_Records'). Rank = transparent score (annex 'Priority_Score'). Scenario date: Mon 9 Nov 2026, 8:00 am.

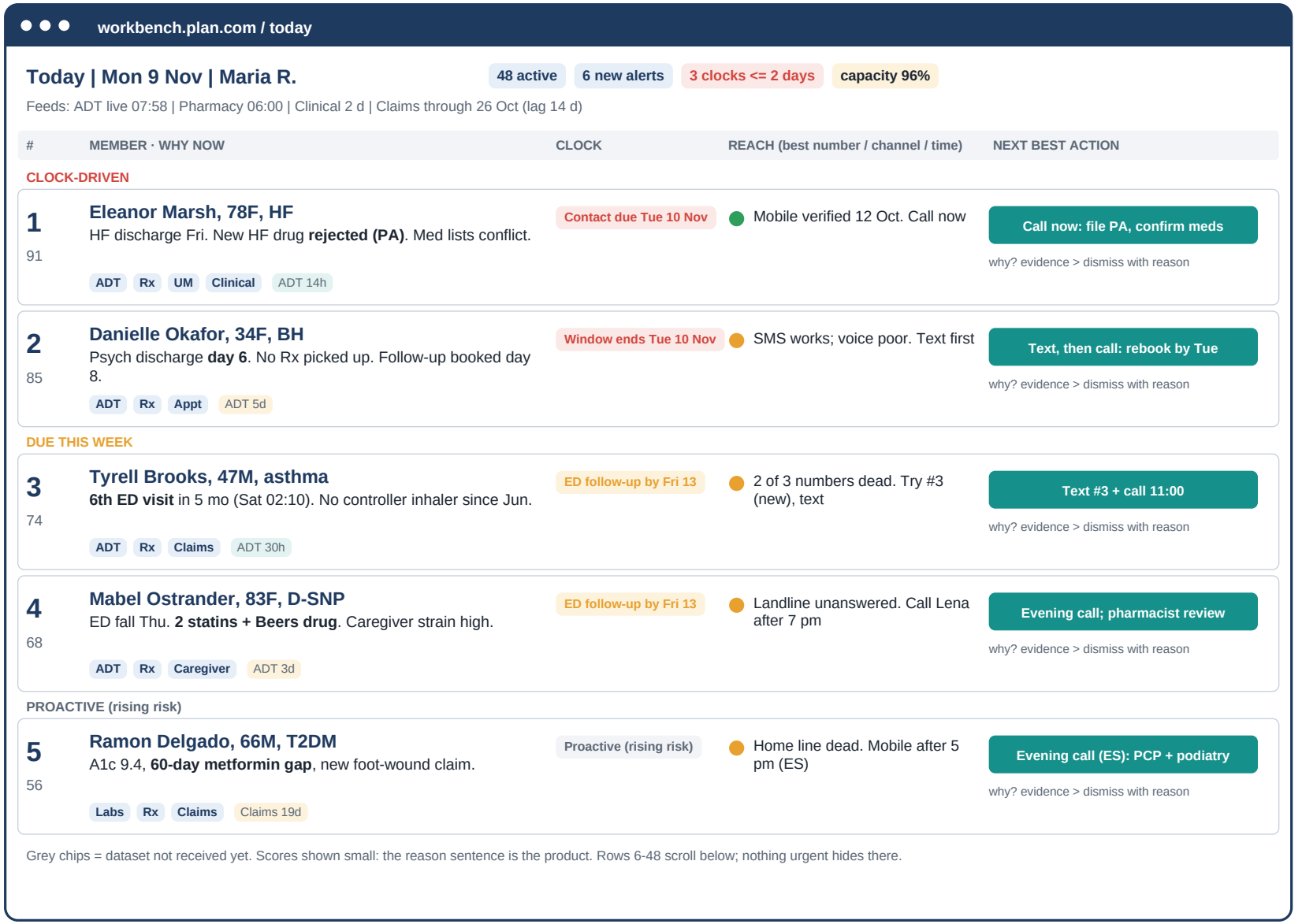
Rank / score	Patient and scenario	Data-sets	Planted imperfections	'8 am Monday': what Maria should see and do
#1 (91)	**P1 Eleanor Marsh** 78F, Medicare Advantage HMO, HFrEF + CKD3 *Fresh heart-failure discharge*	10 datasets	Conflicting med lists (lisinopril vs new drug); ADT says 'Home', summary says 'Home + home health'; inpatient claim not yet received; PA reject	HF discharge Fri; new HF drug rejected for PA; med lists conflict. Call today; submit PA; book PCP <=7d.
#2 (85)	**P4 Danielle Okafor** 34F, ACA Marketplace plan, bipolar II / depression, housing instability *Behavioral health + housing instability*	9 datasets	Stale 'stable housing' screen vs newer note; Rx sent to old-address pharmacy; follow-up booked day 8; conflicting med lists; returned mail; prepaid phone	Psych discharge day 6; no meds; follow-up day 8; housing stale. Text first; rebook <=Tue; Rx nearby; housing referral.
#3 (74)	**P3 Tyrell Brooks** 47M, Medicaid managed care, asthma + HTN *Unreachable frequent ED user*	10 datasets	Name-variant alerts (82% match); 2 of 3 phones dead; ADT 6 ED visits vs claims 4; newest number sits in pharmacy profile; stale 'unable to reach' case	6th ED visit, no controller inhaler since June, 9 failed calls. Try newest pharmacy number + text; Medicaid recert hook.
#4 (68)	**P5 Mabel Ostrander** 83F, dual-eligible D-SNP, T2DM + CKD3b + mild cognitive impairment *Dual-eligible senior with caregiver strain*	9 datasets	Duplicate statins at 2 pharmacies; caregiver number only in free text; landline she cannot hear; ED claim missing; LTSS data monthly (hours under-delivered)	Fall + ED Thu; duplicate statins, Beers-list drug, A1c 6.3 on glipizide; caregiver strain; home-care hours short. Call daughter evening.
#5 (56)	**P2 Ramon Delgado** 66M, Medicare Advantage PPO, T2DM + HTN + CKD2, Spanish-speaking *Rising-risk diabetic who stopped refills*	9 datasets	Duplicate legacy case (M-30027 / M-30072); disconnected home phone; A1c result 3 wks late; foot-wound claim 19 days late; cash fills may hide adherence	A1c 9.4, 60-day metformin gap, foot-wound claim just landed. Evening Spanish call; PCP + podiatry; merge duplicate case.

Why these five. They stress different parts of the design: P1 clock + conflicting meds + blocked drug (Member 360 and queue clock); P2 slow-burn rising risk with uncertain adherence (confidence labels); P3 reachability failure (contact graph); P4 behavioral health + stale SDOH + channel choice (SMS, window logic, consent masking); P5 polypharmacy + caregiver + dual-eligible data lag (multi-source, caregiver as contact).
Score = 30% deadline urgency + 25% acuity + 20% active blocker + 15% time since last real contact + 10% social vulnerability. The score orders the queue; the 'why' text, not the number, is what Maria reads.

All names, member IDs, phone numbers (555-01xx), facilities and results are invented. No real PHI. Score inputs and rationale: annex sheet 'Priority_Score'.

D. Product design (1/3): Today queue answers 'who, why now, what next' above the fold

Wireframe at 8:00 am Monday for Maria; five of her 48 cases shown (the synthetic patients). Ordered by clock first, then score.



How to read it

1. **Above the fold = the day.** Top 5 members, each with a one-line reason and a clock. Grouped by deadline before risk, because a missed 2-business-day window cannot be recovered.
2. **Score is a sort key.** 30% deadline, 25% acuity, 20% active blocker, 15% time since last contact, 10% social vulnerability. Top 3 drivers on hover. Max ~8 'due today' vs her capacity to avoid alert fatigue.
3. **Reach column.** Best number, channel and time from the contact graph, so she never redials a dead number.
4. **Source chips + freshness.** Tap a chip to see the evidence record and as-of date. Feeds banner shows claims lag.
5. **Next best action (NBA).** One per member; a button that starts the call or text and opens the 360.

How NBA is chosen (rules first, model second).
trigger (ADT discharge) > program playbook (HF transition) > check each playbook step against data (fill? visit booked? auth?) > check barriers (phone, SDOH, language) > **pick the earliest unmet step that has a deadline**.
AI assists: writes the 'why now' sentence from structured facts and cites sources. **Nurse:** accepts, snoozes or dismisses with a reason; dismissals retune ranking weekly.

Design basis: transitions-of-care timing (CMS TCM 2 business days / 7-14 day visit; HEDIS TRC, FUH, FMC); alert-fatigue evidence (Ancker et al. 2017). Patient rows are synthetic (annex).

D. Product design (2/3): Member 360 shows one picture, with the age, source and conflicts of every fact

Wireframe for Eleanor Marsh (P1) at 8:05 am. Reading order follows Maria's workflow: situation, what happened when, what is wrong with the meds, what to do.

workbench.plan.com / member / M-30011

Eleanor Marsh 78F | M-30011 | MA-HMO | English | PCP Dr. S. Iyer

HF program | Readmit risk 0.41 (up) | Day 3 after discharge | Contact due Tue 10 Nov

ADT 14h | Clinical 2d | Pharmacy 1d

Claims through 26 Oct (lag 14d)

SITUATION (AI draft, confirm)

Admitted 2-6 Nov, HF exacerbation, EF 25%

ADT | Clinical — Confirm | Edit

New sacubitril/valsartan **not dispensed**: claim rejected, PA required, none filed

Rx | UM — Confirm | Edit

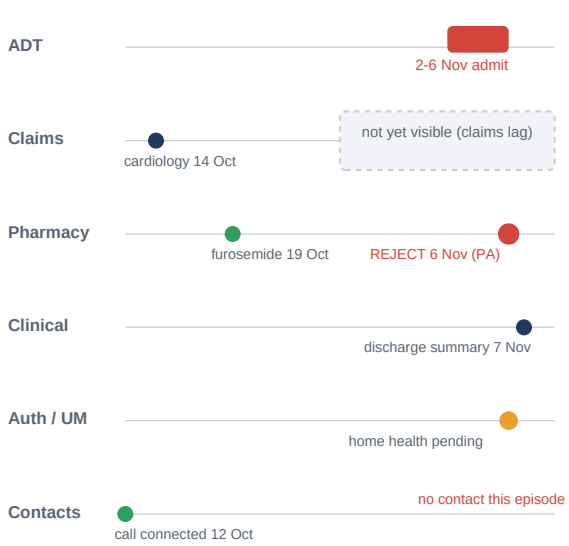
PCP list still shows **lisinopril**; discharge list stopped it

Clinical | HIE — Confirm | Edit

GAPS AND BARRIERS

- No PCP visit booked (cardiology is day 14)
- Home-health auth pending since 6 Nov
- Home scale: unknown, ask
- Husband 84, hearing impaired
- Lives with spouse; transport OK (Mar, stale)

WHAT HAPPENED WHEN (12 Oct to 9 Nov)



Hollow / grey = expected but not received. Absence is never shown as 'no event'.

MEDICATION RECONCILIATION

Drug	Discharge 6 Nov	PCP list 3 Oct	Last fill
Sac/val	24/26 BID (new)	-	REJECTED PA
Lisinopril	STOPPED	10 mg daily	-
Furosemide	40 mg BID	20 mg daily	40 mg, 19 Oct
Carvedilol	12.5 mg BID	12.5 mg BID	12 Sep (gap?)
Spironol.	25 mg (new)	-	none

Red = conflict or blocked, amber = needs check. Both values stay visible with source and date. **Product never picks a winner**; Maria confirms with member and prescriber.
Fill confidence: High (PBM claim). Carvedilol: Medium (discharge supply possible).

NEXT BEST ACTIONS (ordered by deadline; each one-click)

1 File the prior auth

Task to UM with discharge summary attached; ask prescriber about lisinopril overlap.

Create task

2 Call Eleanor now (mobile, verified)

Prompts: weight today? scale at home? symptoms? who picks up meds? which pills in the cabinet?

Start call

3 Book PCP visit by Fri 13 Nov

In-network PCP slots; offer ride; chase home-health auth.

Find slots

How to read it

1. **Header = status and freshness.** Program, risk with trend, clock, and the age of every feed at all times. Claims 'data through' is amber by design.
2. **Situation = AI draft with receipts.** Three lines max, each with source chips. Nurse taps Confirm or Edit; only confirmed lines enter the case record.
3. **Timeline with ghosts.** One lane per dataset. Grey zone shows where claims cannot yet see; hollow markers show expected-but-missing items.
4. **Medication reconciliation.** Source-by-source columns; conflicts coloured; confidence on fills (cash fills may be invisible). No auto-merge: this is Maria's clinical judgment moment.
5. **Actions at the bottom.** Each action is a task with owner and deadline, created in one click; drives the queue.

Design basis: HEDIS TRC (med reconciliation, patient engagement), Part D adherence (PDC); AGS Beers. Clinical statements are decision prompts for the nurse and prescriber, not advice generated by the product.

D. Product design (3/3): Outreach and documentation turn a call into a record, not a second job

Wireframe for Danielle Okafor (P4), day 6 after psychiatric discharge. Left: reach her. Right: document once, confirm once.

workbench.plan.com / member / M-30058 / outreach + documentation

Danielle Okafor 34F | Marketplace Silver | English | Post-psych discharge day 6 | **BH consent: limited sharing (notes masked for non-BH roles)**

OUTREACH (contact graph)

Recommended: text now, call 12:00-14:00. Texting got a reply on 4 Nov; 2 voicemails; prepaid phone often out of minutes. Consent to SMS on file.

#	Number / channel	Source	Evidence
1	555-0161 mobile (prepaid)	Enrollment	2 voicemail; **SMS reply 4 Nov**
2	555-0188 (friend)	Call centre 30 Oct	Unverified; ask consent first
3	Mailing address	Enrollment	Returned 21 Oct: do not use

SMS DRAFT (AI, you review and send)

"Hi Danielle, this is Maria, a nurse from your health plan. I am checking in after your hospital stay. Can I call you today? Reply YES or tell me a better time. Reply STOP to opt out."

Edit | Send

CALL PROMPTS (from the data; order is Maria's)

- Safety check first (follow your protocol)
- Meds: quetiapine + sertraline not picked up. Why? Which pharmacy is near you now?
- Follow-up booked day 8 (11 Nov): can we move it to Tue 10 Nov by video?
- Housing: friend's place per 30 Oct note; confirm current location
- Transport / phone minutes: any barrier?

DOCUMENTATION (live during the call)

BH follow-up assessment: 38 fields



24 pre-filled (confirm) **8 stale** (re-ask) **6 to ask**

Field	Value	State
Discharge date, facility	3 Nov, inpatient psych	Filled: ADT
Discharge meds	sertraline 100, quetiapine 100	Filled: summary
Meds taken since discharge	?	Ask
Housing	'Stable' (Jun) vs 'friend's place' (30 Oct)	Stale: conflict
Phone / SMS consent	Mobile, SMS yes	Filled: confirm
Safety screen result	?	Ask (never auto-filled)

NOTE DRAFT (AI from your quick tags; not saved until you sign)

[Draft] Called member day 6 post-discharge. Reports *[you fill]*. Discharge meds not yet obtained; Rx to be re-sent to *[pharmacy]*. Follow-up moved to *[date]*. Barriers: *[housing, phone minutes]*. Plan: *[next contact]*.

CARE PLAN (auto-updated, confirm)

Goal: attend psychiatry visit by 10 Nov | Barrier: phone, transport | Intervention: telehealth, ride, Rx to nearby pharmacy

One entry writes assessment, plan, call log and audit export.

Review, sign and file (once)

How to read it

1. **Channel, time, number: a recommendation with the reason.** Ranked from source, recency and past success (cross-source: pharmacy, provider, call centre). Dead numbers are demoted automatically.
2. **SMS and call prompts.** Drafted from the data, sent only by the nurse, only with consent on file (TCPA). Prompts follow clinical order; she chooses.
3. **Assessment pre-filled with states.** Green: from records, confirm. Amber: stale or conflicting, re-ask. Red: only the member can answer, never auto-filled (e.g. safety screen).
4. **Note and care plan are drafts.** Structured quick-tags in, narrative out, with blanks where a clinical fact is needed. Nothing is saved until she signs.
5. **Write once, file once.** One sign-off updates assessment, plan, call log and audit form. Removes the same-fact-four-times burden.

Design basis: CMS TCM / HEDIS FUH 7-day follow-up; TCPA consent for SMS; 42 CFR Part 2 and plan BH consent rules (role-based masking); NCQA/URAC case-management documentation expectations.

D. Walkthrough: three patients through the screens, and where AI stops and the nurse starts

Same three screens, three different stories. Each step names the data that makes it possible.

Patient	1 Queue (8:00)	2 Member 360 (8:05)	3 Outreach / documentation (8:15+)	What Maria does differently vs today
P1 Eleanor Marsh HF discharge, conflicting meds	Rank 1 (91). Clock: contact due Tue. Why: HF discharge, new drug rejected (PA).	Rx lane shows REJECT 6 Nov; claims lane greyed (lag). Med table: lisinopril stopped vs still on PCP list.	Mobile verified. Call prompts: weight, scale, meds in cabinet. PA task to UM; PCP visit by Fri 13.	Finds the blocked drug on day 3, not day 7 or at readmission. Flags the overlap to the prescriber instead of discovering it later.
P3 Tyrell Brooks Frequent ED, unreachable	Rank 3 (74). Why: 6th ED visit (ADT), no controller fill. Reach: 2 of 3 numbers dead.	ADT shows 6 ED visits vs claims 4 (match 82% flagged). Contact graph: #3 from pharmacy profile, never tried.	Text #3, call 11:00; ask pharmacy and PCP to relay; Medicaid recertification (30 Nov) as the reason to talk.	Stops redialling dead numbers (9 failed attempts). First connect comes from a number found in another system.
P4 Danielle Okafor BH + housing	Rank 2 (85). Clock: window ends Tue. Reach: SMS works, voice poor.	Appointments lane: follow-up day 8 outside 7-day window. Housing badge: stale screen conflicts with 30 Oct note.	SMS first; assessment 24/38 pre-filled; safety screen left for her; care plan goal 'psychiatry visit by 10 Nov'.	Rebooks inside the window; fixes the pharmacy location; does not re-ask what the plan already knows.

Human-in-the-loop: what is automated, what is suggested, what the nurse decides

Automated (no review needed)	AI-drafted, nurse must confirm	Nurse decides	Never done by the product
Record matching and dedupe; freshness and 'data through' labels; adherence (PDC) maths; contact-number ranking; score calculation; clock calculation.	'Situation' summary; next best action; SMS draft; note draft; assessment pre-fill; phone numbers lifted from free text; merge candidates for duplicate cases.	What the member said; med reconciliation outcome; risk and safety assessment; care plan goals; whether to escalate; closing or transitioning the case.	Diagnose or advise treatment; send messages without consent; auto-resolve conflicting data; write unconfirmed text into the record; auto-close a case.

Trust mechanics: every AI line cites its source records; edits and dismissals are logged and reviewed weekly with a nurse advisory group; the product shows what it does not know ('not received yet', 'confidence: medium') instead of filling gaps; accepted-without-edit rate and override reasons are tracked as product metrics.

Synthetic patients: annex sheet 'Synthetic_Records'. Score values from annex 'Priority_Score'.

E. Prioritize and measure: ship the smallest thing that gives Maria her mornings back, then prove it in a pilot

MVP cut and logic, success metrics with baselines and targets, risks with mitigations, and a pilot / go-to-market sketch.

MVP vs later

MVP (12-16 wks, 1 program)	Phase 2	Later
Queue with clock + reason + score; Member 360 on D1-D5, D7, D9-D12, D21; contact graph; med conflict view; AI summary, note and assessment pre-fill with confirm; SMS with consent	SDOH closed-loop referrals; care gaps (HEDIS); network search + booking; benefits; caregiver profile; scheduling feeds; labs trends	RPM / wearables; Medicaid LTSS feeds; voice-to-note; member app; predictive dialer; multi-language auto-outreach

Prioritisation logic. Rank each capability by (decisions it changes x frequency x clinical risk) / (integration effort x data-quality risk). ADT + pharmacy + discharge summary score highest: they create the clock and explain the event, and the plan already holds them. Scheduling and wearables score low: high effort, patchy data. **Cut line:** if it does not change a decision on Monday at 8 am, it is Phase 2.

Pilot and go-to-market (brownie points)

Buyer. Economic: CMO / VP Care Management; also Stars and Quality lead (TRC, FUH, adherence). Champion: Director of Case Management. Target: regional MA / D-SNP / Medicaid plan, 150-500k members, already ingesting claims and ADT.

Pilot. 12 weeks, one program (transitions of care + HF/COPD), ~25 nurses (~1,200 active cases), shadow 8 nurses for baseline in weeks 1-2, matched control team. Go/no-go gates: prep time -50%, TOC 2-day contact >= 80%, nurse NPS positive.

Pricing signal (assumption). Per-nurse-seat SaaS, ~$400-600/seat/month (~$5-7k/yr) vs ~$110-130k fully loaded RN. A 20% capacity gain on 25 nurses is ~5 FTE (~$0.6M/yr), plus Stars / readmission upside; land with seats, expand with PMPM tiers.

Wk 1-2
Shadow 8 nurses; baseline

Wk 3-4
Connect ADT, Rx, CCD; go live for 5 nurses

Wk 5-12
Scale to 25; weekly review

Wk 12
Readout vs control; go/no-go

Success metrics (baselines are assumptions to be confirmed in pilot week 1)

Metric	Baseline	Pilot target
Prep time per case touch	6.0 min	<= 2.5 min (-58%)
Documentation time per touch	8.0 min	<= 4.5 min (-44%)
Real-conversation share of day	~12% (60 min)	>= 25% (design model: 39%)
Outreach connect rate	~30%	>= 45%
TOC contact within 2 business days	~55%	>= 85%
Active cases per manager (complex)	48	58 (+20%) at same quality
After-hours documentation	~45 min/day	< 10 min/day
30-day readmission, engaged HF/COPD cohort (lagging)	plan baseline	-10% relative
Trust: suggestions accepted / overridden with reason	n/a	>= 70% accepted; overrides reviewed weekly

Risks and mitigations

Risk	Mitigation
Wrong or stale data drives a bad call	Every fact shows source + as-of date; conflicts shown side by side, never silently merged; claims never used as trigger; 'not received yet' state.
Alert fatigue	One next-best-action per member, max ~8 'due today'; snooze with reason; weekly review of dismissed alerts; cap on banners.
Clinician trust / deskilling	AI drafts only; citations on every claim; confirm-before-save; nurse can edit anything; transparent score drivers; nurse advisory panel.
Privacy (BH/SUD, TCPA)	Role-based masking, consent flags, audit log, SMS only with consent.
Integration slips (ADT, scheduling)	MVP runs on ADT + claims + pharmacy + CCD; scheduling feeds are Phase 2.

Sources: BLS / public RN salary ranges (assumed $110-130k loaded); SaaS seat-pricing norms; NCQA HEDIS; CMS Star Ratings. Pricing and ROI are directional, not quotes. Metric details in annex sheet 'Metrics'.